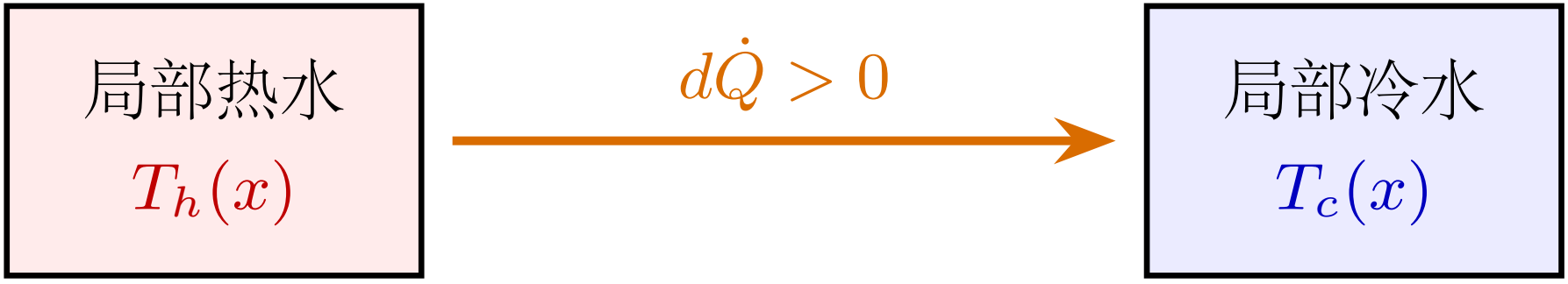

$$d\dot{S}_h = -d\dot{Q}/T_h \qquad d\dot{S}_c = d\dot{Q}/T_c$$

$$d\dot{S}_{\text{gen}} = d\dot{Q}(1/T_c - 1/T_h) > 0$$